

Melanie Fernández

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37-14 91st Street, Jackson Heights, NY 11372

Software engineer and AI researcher graduated in May 2026 with experience building full-stack applications, data platforms, machine learning systems, and AI-assisted development workflows. Passionate about rapidly prototyping products, automating complex processes, and translating ambiguous requirements into deployable software solutions.

TECHNICAL SKILLS

Languages: Python, JavaScript, SQL, Java, C#, C++, R, HTML/CSS, Dart

Backend/APIs: Django, Django REST Framework, FastAPI, Flask, RESTful API Design, JWT Authentication, Flutter

Frontend: JavaScript, HTML/CSS, React.js, D3.js

Databases: SQL, MySQL, SQLite, PostgreSQL, Data Modeling, Query Writing

ML/AI: PyTorch, TensorFlow, scikit-learn, XGBoost, NumPy, pandas, JAX

Dev Tools: Git, Linux/Bash, VS Code, Vim/Neovim, Agile Collaboration, Debugging

EDUCATION

B.A. Computer Science, Minor in Sociology | Connecticut College | New London, CT

Aug 2022 – May 2026

- Focus: Machine Learning and Artificial Intelligence
- Relevant coursework: Data Structures, Algorithms, Computational Intelligence, Database Systems, Computer Networks, Data Visualization, Computer Organization
- Science Leaders Program — honors cohort integrating scientific and social research

EXPERIENCE

Independent Software Developer | Mercury Mental Healthcare | Freelance / Remote

May 2026 – Present

- Developing a browser-based mental health survey platform from client requirements through deployment, rapidly iterating on user experience, data collection workflows, and reporting functionality.
- Debugging frontend and data-flow issues across browser-based deployment environments to ensure stable survey completion and reliable response storage.
- Collaborating directly with stakeholders to refine requirements, prototype features, and deploy updates in response to user feedback.
- Designing secure and user-friendly interfaces optimized for iPad/web deployment while considering HIPAA-compliant data handling practices.

ML Research Engineer | Autonomous Agent Learning Lab, Connecticut College | New London, CT

Jun 2023 – May 2026

- Engineered Python data pipelines to ingest, clean, and transform large structured datasets, reducing preprocessing time by ~35% across multiple ML experiments.
- Designed and trained CNN models on board-state representations for the strategy game Shogi, achieving results published at an international peer-reviewed conference.
- Improved experimental reproducibility by standardizing training workflows and model checkpointing, enabling faster iteration cycles across the research team.

Teaching Assistant — Algorithms & Computational Intelligence | Connecticut College | New London, CT

Sep 2025 – May 2026

- Supported undergraduate students in algorithmic reasoning, computational problem-solving, and machine learning concepts through one-on-one guidance and collaborative instruction.
- Worked closely with faculty to reinforce classroom material and improve student understanding across varying technical backgrounds.

ML Research Fellow | SZTAKI MILAB / AIT Budapest | Budapest, Hungary

Aug 2024 – Dec 2024

- Built end-to-end Python pipelines for cleaning, validation, and transformation of biomedical datasets at scale.
- Implemented KNN imputation and regression models to recover missing values, improving dataset completeness from 62% to 95% and enabling downstream model training.
- Collaborated with computational biology researchers to evaluate model reliability on aging biomarker data, translating domain requirements into ML solutions.

Data Analyst | Connecticut College | New London, CT

Sep 2022 – Dec 2022

- Cleaned, validated, and standardized alumni datasets using SQL and Excel to support institutional reporting and improve long-term data reusability.

PROJECTS & PUBLICATIONS

SupportHub — Full-Stack Ticketing System

2024

- Built a full-stack support platform using Django REST Framework and JWT authentication, enabling role-based workflows for customers, staff, and managers.
- Designed RESTful APIs and frontend interfaces supporting ticket creation, assignment, tracking, and resolution across multiple user roles.

Trader Joe's Retail Database System

2026

- Designed and implemented a normalized MySQL database modeling retail operations across physical and online stores, including inventory, transactions, customers, vendors, and restocking workflows.
- Built command-line workflows supporting customer checkout, cashier operations, inventory updates, and restocking simulations using SQL and Bash.

Interactive Star Data Visualization

2024

- Built an interactive astronomical data explorer using D3.js with dynamic filtering, responsive visualizations, and real-time analysis of large star datasets.

Opening-as-Task Meta Learning for Cross Game Position Evaluation in Chess and Shogi

2026

- Developing an ANIL+MAML framework to transfer policy and value network knowledge between Chess and Shogi, enabling rapid adaptation with minimal retraining.

Pending Publication (SMC): Fernández & O'Connor (2026)

MAGE: MAP-Elites for Alpha Generation

2026

- Applied MAP-Elites and genetic programming to generate diverse and robust alpha trading factors on S&P 500 market data for quantitative finance research.

Pending Publication (CIFE): O'Connor & Fernández (2026)

Shogi Match Prediction — Deep CNN

2023 – 2025

- Built and trained a deep CNN to predict online match outcomes from board-state representations, iterating across multiple model architectures to improve predictive accuracy.
- Extended the model with improved architectures and deeper analysis, accepted as a post-publication extension in *Computational Intelligence*, Springer 2025.
- Nominated for Best Student Paper at IJCCI 2024 in Porto, Portugal.

Published: O'Connor & Fernández, IJCCI 2024 (Porto) · Extended: Computational Intelligence, Springer 2025

Simulation of Speciation and Complex Trait Evolution

2025 – 2026

- Contributed to a multi-species artificial environment simulation modeling evolutionary speciation and complex trait development across agents.

Published: Parker, O'Connor, Nash & Fernández, ICAART 2026 (Marbella, Spain)